concordance=TRUE

POMONA COLLEGE	Environmental Analysis Teaching and Research Laboratory	Date: 6/16/2025	SOP No. X
	Standard Operating Procedure	Title: Nitrate and Ammonium	
	Approved By: Los Huertos	Revised: June 16, 2025	

1. Scope and Application

1.1

1.2 [1]

2. Health and Safety

2.1 [2]

3. Personnel & Training Responsibilities

3.1 [1]
Students using this SOP should be trained for the following SOPsa:

- SOP1
- SOP2

4. Required Materials

4.1 Distilled water

4.2 Calcium Chloride

4.3 Nitrate test strips

4.4 Falcon tubes(volumetrically marked centrifuge tubes?)

4.5 Soil Probe (Auger?)

4.6 Bucket

5. Estimated Time

5.1 This will take XX minutes...

Table 1: Soil Texture Correction Factor

Texture	Moist Soil	Dry Soil
Sand	2.3	2.6
Loam	2.0	2.5
Clay	1.7	2.2

6. Procedure

6.1 Preparation: Add 6 g (1tspn) CaCl2 to 1 gal distilled water (makes 125 tests).

6.2 Fill volumetric container w/ 30 ml solution

6.3 Add soil into container until solution reaches 40 ml

6.4 cap and shake until all soil is well mixed

6.5 allow sample to sit and soil particles to settle (around 10 mins, no more than 1hr)

6.6 Dip test strip in clearer solution near top for one second

6.7 compare strip to standard color chart at 30 and 60 seconds

6.8 Interpretation of N testing quick strips

6.9 Calculate correction factor using soil textures (average values) See Texture Analysis SOP.

6.10 Corrected value = test strip value of N (ppm) / correction factor

6.11 record nitrate ppm and correction factor in field notebook

7. References

7.1 APHA, AWWA. WEF. (2012) Standard Methods for examination of water and wastewater. 22nd American Public Health Association (Eds.). Washington. 1360 pp. (2014).